Swarm Witness Evidence for Agent Authorization

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Independent

With
Apart Research

Track 4, Multi-Agent Oversight and Control

ONE EXECUTION

The allowlist passes. The forbidden write commits.



Same history. Each record answers only its own proposition.

The local control remains supported.

PRIOR ART, ALREADY IN PLACE

BMO

Owned authority
and settlement.

SoundGate

Whether restart preserves
a durable refusal.

EvidenceGraph

A receipt does not establish
event attribution.

COMPARED IN THIS PAPER

CONTRACT

BMO records and a SQLite gate

Separate implementation.
Not an adjudicator.

APPLICATION PACKETS

Qualifier and the invariant checker

Paired methods.
Same 48 packets.

COMPOSITION

Both methods read one adapter

EvidenceGraph records.
Not a new source.

SoundGate is prior art. It is not the paired method on the application packets.

EVIDENCE CHAIN

A later boundary needs its own observation.

Boundary	What it can establish	What it cannot establish alone	Where it is tested
Authority and claim	A scoped one-use claim by the current owner	That the delegated edit applied	BMO and SQLite, 6 cases, 22 steps
Terminal settlement	An applied edit, or a definite no-effect	System safety, or a live external agent	S01-S09, including the initial 2
Effect and warrant	A committed effect, or a justified absence	Safety from a local pass, or from an incomplete ledger	12 histories, viewed as 48 packets
Evidence translation	A classifier conclusion from admitted premises	That the adapter's premises were independently verified	70 of 130 packets

Supported · Refuted · Unresolved · *Unsupported support* · Untested or not carried forward

A later boundary does not revise an earlier count.

NINE SETTLEMENT CASES

Effect-applied is one proposition.

S01

Bound applied records

effect entered

Supported

S02

Bound no-effect records

not entered

Refuted

S03-S09

Refused

terminal evidence invalid

Unresolved

S02 refutes only the effect-applied proposition. The committed claim remains.

Both arms match. Fixture-authored records. Not a live model edit.

Not P_effect. Not P_system. Not part of the six ledger cases.

PUBLISHED APPLICATION QUALIFIER, NOT A BMO RESULT

The replay does not replace Table 2.

HISTORICAL SWARM WITNESS

11 *unsupported supports*

24/24 correct decisive

13/24 correct abstentions

Table 2. The eleven stay.

CURRENT EVALUATOR

0 *unsupported supports*

24/24 correct decisive

24/24 correct abstentions

Table 2b. A later replay.

The sixteen-case challenge is its own denominator.

A true proposition can still be an unwarranted support.

COMPOSITION

The slice already belongs to the corpus.

70-packet slice	inside 130

Derived records are not extra sources.

⊖ RAW, EITHER METHOD

0/36 / 0/26

Assessor and warrant coverage. Abstains.

ADAPTED, EITHER METHOD

36/36 / 26/26

△ Ten *unsupported supports* remain.

Both methods accept the adapter. The ten are the assessor's supports.

Not a Swarm Witness-specific advantage.

Each boundary justifies its own claim, and no later one.

Bounded contract portability is demonstrated across the two implementations.

The *eleven unsupported supports* remain Table 2. The *ten unsupported supports* remain Table 4.

External empirical transfer remains untested.

github.com/instagrim-dev/swarm-witness
Reproduction commands are in evidence/ACCESS.md.